

1. What is data cleaning in Pandas?

Data cleaning in Pandas is the process of identifying and fixing errors, inconsistencies, and missing values in a dataset so that the data is accurate, consistent, and ready for analysis or machine learning.

Common data cleaning tasks in Pandas

1. Handling missing values
2. Removing duplicate data
3. Correcting data types
4. Handling inconsistent data (Fix spelling, case, or formatting issues)

2. What is the difference between loc and iloc?

◆ loc (Label-based indexing)

- Uses row and column labels
- Works with index names and column names
- End index is included
- Can use boolean conditions

◆ iloc (Integer position–based indexing)

- Uses integer positions (0-based index)
- Works like Python list indexing
- End index is excluded
- Does not use labels or conditions directly

◆ loc → **Select by name**

◆ iloc → **Select by position**

`loc()`

```
df.loc[2, 'Age']          # Row with label 2, column 'Age'  
df.loc[1:3, 'Name']      # Rows 1 to 3 (inclusive)  
df.loc[df['Age'] > 30]    # Conditional selection
```

`iloc()`

```
df.iloc[0, 1]            # First row, second column  
df.iloc[0:3, 1:3]        # Rows 0 to 2, columns 1 to 2
```

3. What are missing values and how does Pandas handle them?

Missing values are data entries where no value is recorded for a variable. In Pandas, missing values are usually represented as NaN (Not a Number) or None.

What causes missing values?

- Data not collected or unavailable
- Errors during data entry
- Data loss during merging or transformation
- Optional fields left blank

Pandas handles missing values by:

1. Detecting missing values

```
df.isnull()
df.isnull().sum()
df.notnull()
```

2. Removing missing values

```
df.dropna()           # Remove rows with missing values
df.dropna(axis=1)      # Remove columns with missing values
```

3. Filling missing values

```
df.fillna(0)           # Replace with 0
df.fillna(df.mean())   # Replace with column mean
df.fillna(method='ffill') # Forward fill
df.fillna(method='bfill') # Backward fill
df['Age'].fillna(df['Age'].median(), inplace=True) #Replacing missing values selectively
```

4. What is groupby() and why is it used?

In Pandas, `groupby()` is a powerful function used to split data into groups, apply operations to each group, and combine the results. This is commonly known as the Split–Apply–Combine strategy. It groups rows of a DataFrame based on one or more columns, and then lets you perform aggregation, transformation, or filtering on each group.

Why is groupby() used?

It is used to:

- Summarize data (mean, sum, count, min, max, etc.)
- Analyze patterns within groups
- Perform statistical analysis by category
- Replace complex loops with efficient operations

5. How does Pandas help in data preprocessing for AI models?

Pandas helps in AI data preprocessing by providing efficient tools for data cleaning, transformation, handling missing values, encoding categorical data, and feature selection, ensuring high-quality input for machine learning models.

◆ Handling missing data

Pandas provides functions like `isnull()`, `dropna()`, and `fillna()` to detect, remove, or replace missing values, which is crucial before training AI models.

◆ Data cleaning and correction

It helps in removing duplicates, fixing inconsistent data, correcting data types, and standardizing values using methods like `drop_duplicates()` and `astype()`.

◆ Data transformation and feature engineering

Pandas allows creation of new features, normalization, scaling, and encoding of categorical variables using operations such as column arithmetic and `get_dummies()`.

◆ Filtering and selecting relevant data

It enables easy row and column selection, filtering, and slicing to keep only important features required for AI model training.

◆ Data aggregation and summarization

Using functions like `groupby()`, `mean()`, and `sum()`, Pandas helps summarize and analyze data patterns, aiding better feature understanding before modeling.

